SOC Lab

Step 1

Set up VMware

the VMware is a headache to obtain so i used https://archive.org to get the VMware 17.5 for free ;)
we need:

kali linux : 2gb ram

ubuntu live server : 8gb ram

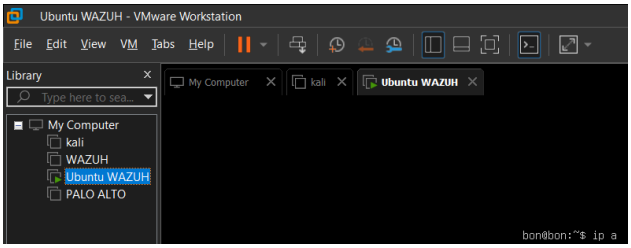
win10 : 4gb ram

> all running as a vm in vmware all of them should be in a NAT adapter configuration

Step 2

setup WAZUH to do so we use a quickstart command from WAZUH documentation for me it failed a couple times until i fixed the fail in the next steps (my fail is a specific kind yours might be different if you even will have an error)

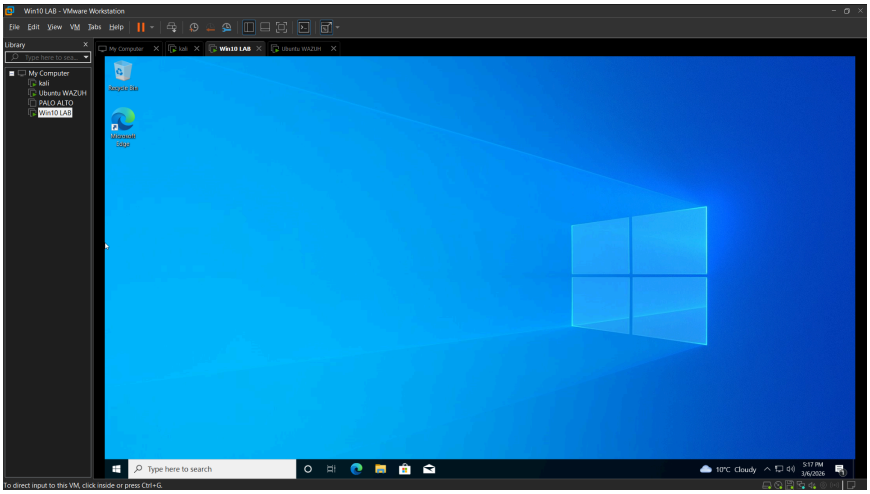
why_WAZUH_installation_was_failing.pdf



> WAZUH credentials:
> username: admin
> password: Salam123?

Step 3

create a Win10 vm



Step 4

install sysmon on it
Sysmon is a background tool that records everything happening on your PC so you can spot security threats.

to do so you go to an official microsoft site and download it from there

afterwards you extract the downloaded file and go to the downloaded directory on powershell and run `sysmon64.exe -i` as administrator

This enables advanced logging.

Step 5

after you access WAZUH on your browser by typing https://<your wazuh ip> and login you go to add new agent and follow the instructions on there

Step 6

after that you create an attacker machine which for us will be a kali linux machine

Step 7

you enable rdp (remote desktop connection) on Win10 mahcine

Step 8

you bruteforce Win10 machine using hydra using command : hydra -t 4 -V -f -l user -P rockyou.txt rdp://Win10 IP

Step 9

you detect it in WAZUH by going to active agents > threat hunting > events and look for failed login attempts

Step 10

now you create a detection rule to trigger an alert if this happens again

by going to server management > rules > add new rules file you create a rule from there

this is what i decided to do which says 5 failed login attempts in 60 seconds then trigger alert

Rules

bruteforce_alert.xml

```xml
<!-- Modify it at your will. -->

<group name="windows, brute_force,">
  <rule id="100001" level="10" frequency="5" timeframe="60">
    <if_matched_sid>60122</if_matched_sid>
    <description>Multiple Windows failed login attempts (Brute Force).</description>
    <mitre>
      <id>T1110</id>
    </mitre>
  </rule>
</group>
```